AFAQ ELEVATORS

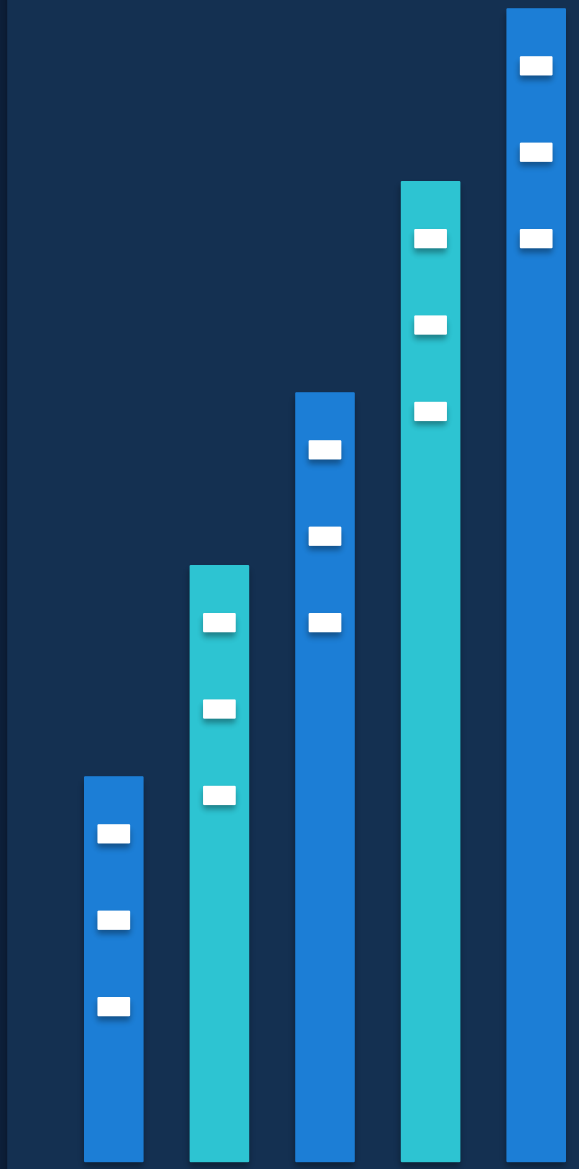
From Operations Data to Better Decisions

A data analytics & database design project

PROJECT DOCUMENTATION • ANALYTICS • DASHBOARD PLAN

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One operational picture. Better decisions.

The project organizes service, maintenance, inventory and financial data into a decision-support view.

WORKBOOK TABLES

15

Operational subject areas

DATA ROWS

514,905

Reported from workbook inventory

CORE DOMAINS

5

Customers • Assets • Service • Supply • Finance

DELIVERY FOCUS

BI-ready

Structured model + KPI dashboard

Business question

Where are service delays, repeat faults, stock risks and financial exposure emerging?

Data approach

Connect related operational records with keys, validate quality, then calculate consistent KPIs.

Expected value

A shared view of performance to support prioritization, planning and follow-up.

Note: row counts describe the supplied workbook; analytical findings must be calculated and validated before presenting them as results.

Why this project matters

Elevator operations create connected records across the asset lifecycle.

01
Install
Projects & elevators

02
Maintain
Contracts & visits

03
Resolve
Faults & parts usage

04
Replenish
Suppliers & inventory

05
Collect
Invoices & payments

Without connected data

Teams may rely on separate spreadsheets and inconsistent definitions.

With a shared model

Records can be traced across assets, service events, parts and money.

Management outcome

Compare performance, spot exceptions and assign follow-up actions.

A broad operational dataset

The workbook covers the main entities needed for a connected elevator-service analysis.

CUSTOMERS & ASSETS

Customers • Elevators • Branches • Employees

SERVICE DELIVERY

Contracts • Projects • Maintenance visits • Faults

SUPPLY CHAIN

Suppliers • Parts • Purchase orders • Inventory transactions • Parts usage

FINANCE

Invoices • Payments

DATA INVENTORY

15

related tables

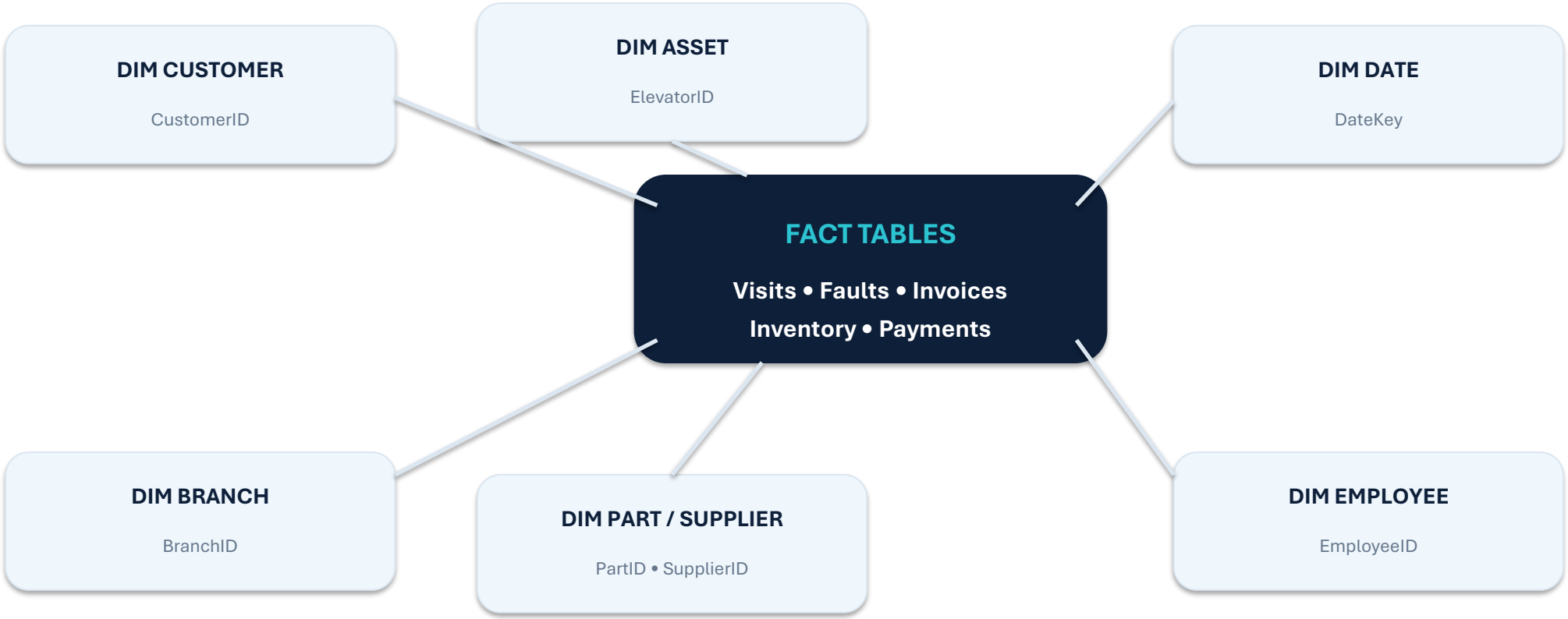
514,905

rows reported across the workbook

Data quality flags and helper columns are present and should be separated from business measures during modeling.

Model for analysis—not just storage

Use stable keys and clear grain to avoid duplicate counting and ambiguous relationships.



Conceptual illustration only. Confirm each relationship by testing key uniqueness and orphan foreign keys before finalizing the ERD.

Trace records end-to-end

Examples of likely joins based on the workbook's shared identifier columns; validate before implementation.

Customer	Elevator	CustomerID	One customer → multiple elevators
Elevator	Maintenance visit	ElevatorID	One elevator → many visits
Visit	Fault	VisitID	A visit may be linked to one or more faults
Supplier / Part	Purchase order	SupplierID • PartID	Orders connect procurement to parts
Invoice	Payment	InvoiceID	An invoice may have multiple payments

Validation checklist: PK uniqueness • FK orphan checks • expected cardinality • date consistency • duplicate business records

Quality checks are part of the solution

The workbook includes QF_ quality flags and AN_ analysis/helper fields; these should be documented, not mistaken for raw facts.

Referential integrity

Missing or unmatched customer, elevator, visit, part or invoice references

Time logic

Events after snapshot date, work before hire date, or inconsistent lifecycle dates

Operational logic

Hydraulic faults on non-hydraulic assets, repeat-fault flags, duplicate issue records

Financial logic

Invoice margin reliability, recalculated settlement status, refunds and tiny payments

Recommended process

Profile → Validate → Resolve → Document

Measure what managers can act on

Proposed KPIs: calculate them from validated records and agree on definitions with the business.

SERVICE

First-time fix rate

Faults resolved without a repeat repair

RELIABILITY

Downtime hours

Total and average downtime by asset / fault category

CONTRACTS

Renewal coverage

Contracts due for renewal and active contract value

SUPPLY

Stockout / reorder risk

Parts below reorder level and shortages linked to service

PROJECTS

Schedule variance

Actual completion vs planned end date

FINANCE

Gross margin & overdue

Reliable invoice margin, outstanding balance and aging

Avoid publishing numeric KPI results until formulas, exclusions and date boundaries are verified.

A manager-friendly dashboard layout

Illustrative wireframe—replace with actual Power BI visuals after the data model is validated.



A clear path from workbook to insight

The order matters: data validation comes before dashboard publishing.

01 Profile

Inspect columns, data types, missing values, duplicates

02 Model

Confirm PK/FK, grain, cardinality and date dimensions

03 Transform

Clean records and separate quality flags / helper fields

04 Analyze

Define KPI formulas and validate business rules

05 Visualize

Build pages, filters, drill-through and exception views

06 Deliver

Document assumptions, record demo and publish repository

Suggested hand-off: validated data model + data dictionary + SQL/Python assets + Power BI report + short demo video.

Protect trust in the numbers

The biggest risk is a polished dashboard built on unvalidated relationships or inconsistent business definitions.

Duplicate counting	Validate grain and many-to-many joins before aggregation.
Unmatched keys	Run orphan checks and report exception counts.
Misleading margin	Use only reliable invoice costs and documented exclusions.
Unclear snapshot dates	Define the reporting cutoff and separate future-dated records.
Unproven findings	Label proposed KPIs separately from measured results.

Governance rule: every KPI needs a definition, owner, refresh cadence and validation check.

THE TAKEAWAY

Connect the data.
Validate the story.
Act on the insight.

AFAQ Elevators analytics blueprint

Reliable

Trusted keys & quality checks

Actionable

Operational KPIs & exception views

Scalable

A repeatable model for reporting

Thank you